Landau-de Gennes theory in three dimensions

Complex disordered systems: Liquid crystals

For a uniaxial nematic in three dimensions, the alignment tensor can be written as

$$\mathbf{Q} = \mathcal{S}\left(\mathbf{n}\otimes\mathbf{n} - \tfrac{1}{3}\mathbf{I}\right),$$

where $\mathbf{n}$ is the director and $\mathcal{S}$ the scalar order parameter.

1. Choosing the z axis along $\mathbf{n}$, write $\mathbf{Q}$ explicitly and show that $\operatorname{Tr}\mathbf{Q} = 0$, $\operatorname{Tr}\mathbf{Q}^2 = \frac{2}{3}\mathcal{S}^2$ and $\operatorname{Tr}\mathbf{Q}^3 = \frac{2}{9}\mathcal{S}^3$.
2. Hence show that the Landau-de Gennes free energy density

   $$f = f_0 + \frac{A}{2}\operatorname{Tr}\mathbf{Q}^2 - \frac{B}{3}\operatorname{Tr}\mathbf{Q}^3 + \frac{C}{4}\left(\operatorname{Tr}\mathbf{Q}^2\right)^2$$

   becomes

   $$f - f_0 = \alpha\mathcal{S}^2 - \beta\mathcal{S}^3 + \gamma\mathcal{S}^4,$$

   and give α, β and γ in terms of A, B and C.
3. Find the stationary points of f with respect to $\mathcal{S}$.
4. Sketch $f(\mathcal{S})$ for decreasing values of α, keeping $\beta, \gamma > 0$ fixed. At the transition the isotropic ($\mathcal{S} = 0$) and nematic minima have the same free energy. Show that this happens for $\alpha = \beta^2/4\gamma$, with $\mathcal{S}_{NI} = \beta/2\gamma$. What kind of transition is this, and which term is responsible?

Solution

1. With the z axis along $\mathbf{n}$, the only non-zero entry of $\mathbf{n}\otimes\mathbf{n}$ is $n_z n_z = 1$, so

   $$\mathbf{Q} = \mathcal{S}\begin{pmatrix} -\frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & \frac{2}{3} \end{pmatrix},$$

   which is traceless by construction. Since $\mathbf{Q}$ is diagonal, its powers are obtained by raising the diagonal elements to the same power:

   $$\operatorname{Tr}\mathbf{Q}^2 = \mathcal{S}^2\left(\tfrac{1}{9} + \tfrac{1}{9} + \tfrac{4}{9}\right) = \tfrac{2}{3}\mathcal{S}^2, \qquad \operatorname{Tr}\mathbf{Q}^3 = \mathcal{S}^3\left(-\tfrac{1}{27} - \tfrac{1}{27} + \tfrac{8}{27}\right) = \tfrac{2}{9}\mathcal{S}^3.$$

   Traces are invariant under rotations, so the result does not depend on our choice of axes.
2. Substituting the traces

   $$f - f_0 = \frac{A}{3}\mathcal{S}^2 - \frac{2B}{27}\mathcal{S}^3 + \frac{C}{9}\mathcal{S}^4,$$

   so $\alpha = A/3$, $\beta = 2B/27$ and $\gamma = C/9$. With $A = a(T - T^*)$ as in the notes, α is the coefficient that changes sign with temperature.
3. Set $df/d\mathcal{S} = 0$:

   $$2\alpha\mathcal{S} - 3\beta\mathcal{S}^2 + 4\gamma\mathcal{S}^3 = \mathcal{S}\left(2\alpha - 3\beta\mathcal{S} + 4\gamma\mathcal{S}^2\right) = 0.$$

   So either $\mathcal{S} = 0$ (isotropic) or

   $$\mathcal{S}_\pm = \frac{3\beta \pm \sqrt{9\beta^2 - 32\alpha\gamma}}{8\gamma},$$

which only exist for $9\beta^2 \geq 32\alpha\gamma$. Since $f''(0) = 2\alpha$, $\mathcal{S} = 0$ is a minimum for $\alpha > 0$ and becomes unstable for $\alpha < 0$, i.e. below T^*: T^* is the spinodal of the isotropic phase. When they exist, $\mathcal{S}_+$ is the nematic minimum and $\mathcal{S}_-$ the maximum (the barrier) in between.

4. As α decreases, we go from a single minimum at $\mathcal{S} = 0$, to a second (metastable) minimum at $\mathcal{S}_+$, to two minima of equal depth (the transition), to a nematic minimum lower than the isotropic one, until $\mathcal{S} = 0$ becomes unstable at $\alpha = 0$.

At the transition the nematic minimum is stationary and has the same free energy as the isotropic phase, $f(\mathcal{S}) = f(0)$ and $f'(\mathcal{S}) = 0$. Dividing by $\mathcal{S}^2$ and $\mathcal{S}$ respectively:

$$\alpha - \beta\mathcal{S} + \gamma\mathcal{S}^2 = 0, \qquad 2\alpha - 3\beta\mathcal{S} + 4\gamma\mathcal{S}^2 = 0.$$

Subtracting twice the first from the second gives $-\beta\mathcal{S} + 2\gamma\mathcal{S}^2 = 0$, so

$$\mathcal{S}_{NI} = \frac{\beta}{2\gamma},$$

and back into the first equation

$$\alpha = \beta\mathcal{S}_{NI} - \gamma\mathcal{S}_{NI}^2 = \frac{\beta^2}{4\gamma}.$$

The order parameter jumps from zero to a finite value: the transition is first order. This is due to the cubic term $-\beta\mathcal{S}^3$, which allows a nematic minimum separated from $\mathcal{S} = 0$ by a barrier.
With the coefficients of the notes we recover $T_{NI} = T^* + b^2/27ac$ and $\mathcal{S}_{NI} = b/3c$. Since $T_{NI} > T^*$, between T_{NI} and T^* the isotropic phase can be supercooled.